## Gas, totals and mass flow

Cold gas (stations 0-3, 13-19): $\gamma_c = 1.4$, $cp_c = 1004.5$ J/(kg K), $R_c = 287$ J/(kg K). Hot gas (4-9): $\gamma_t = 1.3$, $cp_t = 1239$ J/(kg K), $R_t = cp_t\,(\gamma_t - 1)/\gamma_t = 285.9$ J/(kg K). $R_t$ applies at hot stations.

$$P = \rho R T, \quad R = c_p\frac{\gamma-1}{\gamma}, \quad a = \sqrt{\gamma R T}, \quad \rho_1 v_1 A_1 = \rho_2 v_2 A_2$$

$$\dot{Q} - \dot{W} = \dot{m}\left[\left(h + \tfrac{v^2}{2}\right)_{out} - \left(h + \tfrac{v^2}{2}\right)_{in}\right]$$

W > 0 extracted (turbine), W < 0 delivered (compressor)

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{P_2}{P_1}, \qquad \left.\frac{T_2}{T_1}\right|_s = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

$$T_t = T + \frac{v^2}{2c_p}, \quad \frac{T_t}{T} = 1 + \frac{\gamma-1}{2}M^2, \quad \frac{P_t}{P} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}}$$

values in the isentropic flow tables (page 4)

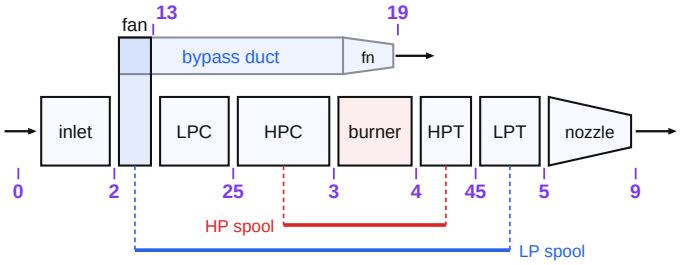
$$\frac{P_{t2}}{P_{t1}} = e^{-\Delta s/R}$$

adiabatic inlet: $\tau_d = 1$ but $\pi_d < 1$; normal shock keeps Tt, loses Pt

$$\dot{m} = \frac{P_t A\,\mathrm{MFP}(M)}{\sqrt{R T_t}}, \qquad \mathrm{MFP} = \sqrt{\gamma} M \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

MFP max at M = 1: 0.6673 (γ 1.3), 0.6847 (γ 1.4); oblique flow: A cos(α); Pt in Pa

## Station numbering (AS755)




two-spool separate-exhaust turbofan (fn = fan nozzle); turbojet: 0-2-3-4-5-9; mixed exhaust: core 6 + bypass 16 → 6A, afterburner exit 7, nozzle throat 8

| Stn | Location | Stn | Location |
|---|---|---|---|
| 0 | freestream | 4 | burner exit / HPT vanes |
| 1 | inlet entry | 45 | HPT exit / LPT entry |
| 2 | inlet exit / fan face | 5 | LPT exit |
| 13 | fan exit | 6, 16 | mixer entry: core, fan duct |
| 25 | LPC exit / HPC entry | 6A, 7 | mixer exit, afterburner exit |
| 3 | HPC exit / burner entry | 8, 9, 19 | throat, core exit, fan exit |

## Cycle notation and components

$$\tau_r = 1 + \frac{\gamma_c - 1}{2}M_0^2, \quad \pi_r = \tau_r^{\frac{\gamma_c}{\gamma_c - 1}}, \quad \tau_\lambda = \frac{c_{pt} T_{t4}}{c_{pc} T_0}, \quad f = \frac{\dot{m}_f}{\dot{m}_3}$$

$$\pi_d = \pi_{d,\max}\eta_r, \quad \eta_r = 1\ (M_0 \le 1), \quad 1 - 0.075(M_0 - 1)^{1.35}\ (1 < M_0 < 5),$$

$$\eta_d = \frac{(P_{t2}/P_0)^{\frac{\gamma-1}{\gamma}} - 1}{T_{t0}/T_0 - 1}$$

$$\tau_c = \pi_c^{\frac{\gamma_c - 1}{\gamma_c \varepsilon_c}} = 1 + \frac{\pi_c^{(\gamma_c-1)/\gamma_c} - 1}{\eta_c}, \qquad \dot{W}_c = \dot{m} c_{pc}(T_{t3} - T_{t2})$$

$$\varepsilon = \frac{\gamma-1}{\gamma}: \ \eta_c = \frac{\pi_c^\varepsilon - 1}{\tau_c - 1},\ e_c = \frac{\varepsilon \ln\pi_c}{\ln\tau_c}; \qquad \eta_t = \frac{1-\tau_t}{1-\pi_t^\varepsilon},\ e_t = \frac{\ln\tau_t}{\varepsilon\ln\pi_t}$$

e divides for a compressor (τ = π^(ε/e)), multiplies for a turbine (τ = π^(ε e))

$$f = \frac{c_{pt}T_{t4} - c_{pc}T_{t3}}{\eta_b h_{PR} - c_{pt}T_{t4}} = \frac{\tau_\lambda - \tau_r\tau_c}{\eta_b h_{PR}/(c_{pc}T_0) - \tau_\lambda}, \qquad \pi_b \approx 1 - \varepsilon\frac{\gamma}{2}M_b^2$$

jet fuel $h_{PR} \approx 42.5$-$42.8$ MJ/kg; stoichiometric $f \approx 0.068$

$$\tau_t = 1 - \frac{\tau_r(\tau_c - 1)}{\eta_m(1+f)\tau_\lambda}, \qquad \pi_t = \tau_t^{\frac{\gamma_t}{(\gamma_t-1)\varepsilon_t}} = \left[1 - \frac{1-\tau_t}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t - 1}}$$

## Nozzles

$$T_{t9} = T_{t8} = T_{t5}, \quad \pi_n = \frac{P_{t9}}{P_{t5}}, \quad \eta_n = \frac{T_{t5} - T_9}{T_{t5} - T_{9s}} = \frac{1 - T_9/T_{t5}}{1 - (P_9/P_{t5})^{(\gamma_t-1)/\gamma_t}}$$

stations: 8 throat, 9 exit (same plane in a convergent nozzle); $\eta_n$ is referred to Pt5, $\pi_n$ to Pt9

$$\pi_n = \frac{P_9}{P_{t5}}\left(\frac{T_{t5}}{T_9}\right)^{\frac{\gamma_t}{\gamma_t - 1}}, \qquad C_V = \frac{v_9}{v_{9s}} = \sqrt{\eta_n}$$

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9}\pi_r\pi_d\pi_c\pi_b\pi_t\pi_n, \qquad \frac{P_{t19}}{P_{19}} = \frac{P_0}{P_{19}}\pi_r\pi_d\pi_f\pi_{fn}, \qquad \pi_{fn} = \frac{P_{t19}}{P_{t13}}$$

two spools: $\pi_c = \pi_{cL}\,\pi_{cH}$, $\pi_t = \pi_{tH}\,\pi_{tL}$; fan stream: Tt19 = Tt13, $\gamma_c$, $R_c$

$$M_9 = \sqrt{\frac{2}{\gamma-1}\left[\left(\frac{P_{t9}}{P_9}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]}, \quad T_9 = \frac{T_{t9}}{1 + \frac{\gamma-1}{2}M_9^2}, \quad v_9 = M_9\sqrt{\gamma R T_9}$$

$$T_9 = T_{t5}\left[1 - \eta_n\left(1 - \left(\frac{P_9}{P_{t5}}\right)^{\frac{\gamma-1}{\gamma}}\right)\right],$$

$$v_9 = \sqrt{2c_p(T_{t5} - T_9)}, \quad M_9 = \sqrt{\frac{2}{\gamma-1}\left(\frac{T_{t9}}{T_9} - 1\right)}$$

$$\frac{T^*}{T_t} = \frac{2}{\gamma+1}, \quad \frac{P_t}{P^*} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}}, \quad v^* = \sqrt{\frac{2\gamma}{\gamma+1}RT_t};$$

$$\eta_n:\ P^* = P_{t5}\left[1 - \frac{1 - T^*/T_t}{\eta_n}\right]^{\frac{\gamma}{\gamma-1}}$$

T*/Tt = 0.8696 (γ 1.3), 0.8333 (γ 1.4); Pt/P* = 1.832 (γ 1.3), 1.893 (γ 1.4)

$$\text{convergent: } P^* \ge P_0:\ M_9 = 1,\ P_9 = P^*,\ T_9 = T^*;$$

$$P^* < P_0:\ P_9 = P_0,\ M_9 < 1$$

isentropic: choked for Pt9/P0 ≥ 1.832 (γ 1.3), 1.893 (γ 1.4); the exit static pressure cannot fall below P*

$$\dot{m} = \frac{P_{t8}A_8\,\mathrm{MFP}(1)}{\sqrt{RT_{t8}}}, \quad A_9 = \frac{\dot{m}_9 R T_9}{P_9 v_9} = \frac{\dot{m}_9\sqrt{RT_{t9}}}{P_{t9}\,\mathrm{MFP}(M_9)}, \quad \frac{A_9}{A_8} = \frac{P_{t8}}{P_{t9}}\frac{\mathrm{MFP}(1)}{\mathrm{MFP}(M_9)}$$

isentropic divergent part: A9/A8 = A/A*(M9) (table); MFP(1) = 0.6673 (γ 1.3), 0.6847 (γ 1.4)

$$F_g = \dot{m}_9 v_9 + (P_9 - P_0)A_9, \qquad dF_g = (P_9 - P_0)\,dA_9 \ \Rightarrow\ F_{g,\max}\text{ at } P_9 = P_0$$

P9 > P0 under-expanded, P9 < P0 over-expanded; matched and isentropic: v9 = sqrt(2 cp Tt9 [1 - (P0/Pt9)^((γ-1)/γ)])

Convergent nozzles: subsonic civil engines and fan streams (the fan nozzle is often unchoked at cruise). C-D nozzles: high nozzle pressure ratio (supersonic flight, afterburning), usually variable area.

## Thrust and performance

$$F = \dot{m}_9 v_9 - \dot{m}_0 v_0 + (P_9 - P_0)A_9, \qquad A_9 = \frac{\dot{m}_9 R T_9}{P_9 v_9}, \qquad I = PA(1 + \gamma M^2)$$

two streams: extra term $\dot{m}_{19}$ v19 + (P19 - P0) A19; force on a component = I_in - I_out

$$F_{inst} = F - D_{nac} - D_{add}, \qquad F_{adim} = \frac{F}{a_0\dot{m}_0}, \qquad I_{sp} = \frac{F}{\dot{m}_f g} = \frac{1}{g\,\mathrm{TSFC}}$$

$$\mathrm{TSFC} = \frac{\dot{m}_f}{F}, \quad \eta_T = \frac{\Delta E_k}{\dot{m}_f h_{PR}}, \quad \eta_P = \frac{F v_0}{\Delta E_k}, \quad \eta_O = \eta_T\eta_P = \frac{v_0}{h_{PR}\,\mathrm{TSFC}}$$

TSFC in mg/(N s) = 1e-6 kg/(N s); uninstalled F in the efficiencies

$$\Delta E_k = \tfrac{1}{2}\left(\dot{m}_9 v_9^2 + \dot{m}_{19}v_{19}^2 - \dot{m}_0 v_0^2\right), \qquad \eta_P \approx \frac{2}{1 + v_9/v_0}$$

$\eta_P$ includes the pressure thrust; static (v0 = 0): $\eta_P = \eta_O = 0$

$$\frac{T_9}{T_0} = \frac{\tau_\lambda\tau_t c_{pc}/c_{pt}}{(P_{t9}/P_9)^{(\gamma_t-1)/\gamma_t}}, \qquad \frac{v_9}{a_0} = M_9\sqrt{\frac{\gamma_t R_t T_9}{\gamma_c R_c T_0}}$$

$$\frac{F}{\dot{m}_0} = a_0\left[(1+f)\frac{v_9}{a_0} - M_0 + (1+f)\frac{R_t}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1 - P_0/P_9}{\gamma_c}\right], \qquad \mathrm{TSFC} = \frac{f}{F/\dot{m}_0}$$

$$\tau_{c,\mathrm{opt}} = \frac{\sqrt{\tau_\lambda}}{\tau_r}$$

raising Tt4 and $\pi_c$ raises $\eta_T$ but also v9, which lowers $\eta_P$: the turbofan breaks that coupling

## Turbofan and two spools

$$\alpha = \frac{\dot{m}_{sec}}{\dot{m}_{core}}, \quad \dot{m}_{core} = \frac{\dot{m}_0}{1+\alpha}, \quad \dot{m}_f = f\dot{m}_{core}, \quad \mathrm{TSFC} = \frac{f}{(1+\alpha)F/\dot{m}_0}$$

$$\dot{W}_{HPC} = \eta_{m,HP}(1+f)\dot{m}_{core}c_{pt}(T_{t4} - T_{t45})$$

$$\dot{W}_{fan} + \dot{W}_{LPC} = \eta_{m,LP}(1+f)\dot{m}_{core}c_{pt}(T_{t45} - T_{t5}),$$

$$\dot{W}_{fan} = \dot{m}_0 c_{pc}(T_{t13} - T_{t2}), \quad \dot{W}_{LPC} = \dot{m}_{core}c_{pc}(T_{t25} - T_{t13})$$

$$\mathrm{OPR} = \pi_f \pi_{LPC} \pi_{HPC}, \quad T_{t,out} = T_{t,in}\left[1 + \frac{\pi^{\varepsilon} - 1}{\eta}\right],$$

$$\frac{P_{t,out}}{P_{t,in}} = \left[1 - \frac{1 - T_{t,out}/T_{t,in}}{\eta_t}\right]^{\frac{\gamma_t}{\gamma_t - 1}}$$

$$F = \dot{m}_9 v_9 + \dot{m}_{19} v_{19} - \dot{m}_0 v_0 + (P_9 - P_0)A_9 + (P_{19} - P_0)A_{19},$$

$$FR = \frac{F_{core}/\dot{m}_{core}}{F_{sec}/\dot{m}_{sec}}$$

$\dot{m}_9$ = (1+f) $\dot{m}_{core}$, $\dot{m}_{19}$ = $\dot{m}_{sec}$

Per unit $\dot{m}_0$, core-flow quantities carry the factor 1/(1+α).

## Mixed exhaust, mixer and afterburner

Long-duct turbofan: fan duct $\pi_{fd}$ = Pt16/Pt13 = 0.98-0.99, $\tau_{fd}$ = 1. Kutta condition at the splitter: P6 = P16; design approximation Pt6 ≈ Pt16 closes the cycle, with α set by the pressure match.

$$\pi_t = \frac{\pi_{fd}\pi_f}{\pi_b \pi_c}, \quad \tau_t = \pi_t^{\frac{(\gamma_t - 1)e_t}{\gamma_t}}, \quad \alpha = \frac{\eta_m(1+f)\tau_\lambda(1-\tau_t) - \tau_r(\tau_c - 1)}{\tau_r(\tau_f - 1)}$$

a physical α (≥ 0) exists only for compatible $\pi_f$, $\pi_c$, Tt4; f is independent of α

$$\frac{P_{t6}}{P_0} = \pi_r \pi_d \pi_c \pi_b \pi_t, \quad \frac{T_{t6}}{T_0} = \tau_\lambda \tau_t \frac{c_{pc}}{c_{pt}}; \qquad \frac{P_{t16}}{P_0} = \pi_r \pi_d \pi_f \pi_{fd}, \quad \frac{T_{t16}}{T_0} = \tau_r \tau_f$$

$$\alpha' = \frac{\dot{m}_{16}}{\dot{m}_6} = \frac{\alpha}{1+f}, \quad c_{p6A} = \frac{c_{p6} + \alpha' c_{p16}}{1 + \alpha'},$$

$$R_{6A} = \frac{R_6 + \alpha' R_{16}}{1 + \alpha'}, \quad \gamma_{6A} = \frac{c_{p6A}}{c_{p6A} - R_{6A}}$$

cp and R are mass-averaged; γ follows from them

$$c_{p6A} T_{t6A} = \frac{c_{p6} T_{t6} + \alpha' c_{p16} T_{t16}}{1 + \alpha'},$$

$$M_{16} = \sqrt{\frac{2}{\gamma_{16} - 1}\left[\left(\frac{P_{t16}}{P_{t6}}\left(1 + \frac{\gamma_6 - 1}{2}M_6^2\right)^{\frac{\gamma_6}{\gamma_6 - 1}}\right)^{\frac{\gamma_{16} - 1}{\gamma_{16}}} - 1\right]}$$

M6 = 0.4-0.5 is a design choice

$$\frac{A_{16}}{A_6} = \alpha' \sqrt{\frac{R_{16} T_{t16}}{R_6 T_{t6}}} \frac{P_{t6}}{P_{t16}} \frac{\mathrm{MFP}(\gamma_6, M_6)}{\mathrm{MFP}(\gamma_{16}, M_{16})}, \qquad A_{6A} = A_6 + A_{16}$$

$$P_{6A} A_{6A} (1 + \gamma_{6A} M_{6A}^2) = P_6 A_6 \left[(1 + \gamma_6 M_6^2) + \frac{A_{16}}{A_6}(1 + \gamma_{16} M_{16}^2)\right]$$

implicit in M6A (subsonic root)

$$\pi_M = \frac{P_{t6A}}{P_{t6}} = \pi_{M\,\max}(1 + \alpha')\sqrt{\frac{R_{6A} T_{t6A}}{R_6 T_{t6}}} \frac{A_6}{A_{6A}} \frac{\mathrm{MFP}(\gamma_6, M_6)}{\mathrm{MFP}(\gamma_{6A}, M_{6A})}$$

$\pi_{M,max}$ ≈ 0.97 (friction); the rest is mixing loss

$$f_{AB} = (1 + f + \alpha)\frac{c_{pAB} T_{t7} - c_{p6A} T_{t6A}}{\eta_{AB} h_{PR} - c_{pAB} T_{t7}}, \qquad \pi_{AB} = \frac{P_{t7}}{P_{t6A}}, \quad f_{tot} = f + f_{AB}$$

fuel fractions referred to the core flow; turbojet: α = 0 and station 6 for 6A; dry: $\pi_{AB} = \tau_{AB}$ = 1, f_AB = 0

$$\mathrm{MFP}(\gamma_7, M_7) = \frac{1 + \alpha + f_{tot}}{1 + \alpha + f} \mathrm{MFP}(\gamma_{6A}, M_{6A}) \sqrt{\frac{R_7 T_{t7}}{R_{6A} T_{t6A}}} \frac{P_{t6A}}{P_{t7}}$$

thermal choking if the right side > MFP(1) = 0.667 (γ 1.3); afterburner inlet M typically 0.2-0.3

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9} \pi_r \pi_d \pi_c \pi_b \pi_t \pi_M \pi_{AB} \pi_n, \qquad \mathrm{TSFC} = \frac{f_{tot}}{(1 + \alpha)F/\dot{m}_0}$$

$$\frac{F}{\dot{m}_0} = \frac{a_0}{1 + \alpha}\left[(1 + f_{tot} + \alpha)\frac{v_9}{a_0} - (1 + \alpha)M_0 + (1 + f_{tot} + \alpha)\frac{R_{6A}}{R_c}\frac{T_9/T_0}{v_9/a_0}\frac{1 - P_0/P_9}{\gamma_c}\right]$$

lighting the afterburner ≈ doubles TSFC for ≈ 50 % more thrust (v9 grows ≈ sqrt(Tt7))

$$\frac{F_{g,mixed}}{F_{g,unmixed}} = \frac{\sqrt{(1 + \alpha)(T_h/T_c + \alpha)}}{\sqrt{T_h/T_c} + \alpha}$$

2-7 %, largest near α ≈ 2 and for hot cores; high-bypass civil engines skip it for duct weight and drag

## Turboshaft

$$\tau_{tH} = 1 - \frac{\tau_r(\tau_c - 1)}{\eta_{mH}(1 + f)\tau_\lambda}, \qquad \frac{\dot{W}_{shaft}}{\dot{m}_0} = \eta_{mL}(1 + f) c_{pc} T_0 \tau_\lambda \tau_{tH}(1 - \tau_{tL})$$

$$\dot{W}_{eq} = \dot{W}_{shaft} + F v_0, \quad \mathrm{PSFC} = \frac{f}{\dot{W}_{shaft}/\dot{m}_0}, \quad \eta_T = \frac{\dot{W}_{shaft} + \Delta E_k}{\dot{m}_f h_{PR}}$$

residual jet: Pt9/P9 = (P0/P9) $\pi_r$ $\pi_d$ $\pi_c$ $\pi_b$ $\pi_{tH}$ $\pi_{tL}$ $\pi_n$; $\eta_P$ not meaningful without the rotor

## Propellers, turboprop, open rotor

$$v_p = \frac{v_0 + v_e}{2}, \quad a = \frac{v_p - v_0}{v_0}, \quad F = 2\dot{m} a v_0,$$

$$\dot{m} = \rho_0 A_D v_p, \quad \dot{W}_D = F v_p, \quad \eta_P = \frac{1}{1 + a}$$

$$v_p = \frac{v_0 + \sqrt{v_0^2 + 2F/(\rho_0 A_D)}}{2}, \quad \eta_{prop} = \eta_L \eta_P,$$

$$\eta_L = \frac{\dot{W}_D}{\dot{W}_{prop}} = \eta_{prop}(1 + a) \le 1$$

$\eta_L \le 1$ bounds $\eta_{prop}$ for a given disc; v_prop = ve = 2 vp - v0

$$\eta_{prop} = 10 M_0 \eta_{\max}\ (M_0 \le 0.1), \quad \eta_{\max}\ (0.1 < M_0 \le 0.7),$$

$$\eta_{\max}\left[1 - \frac{M_0 - 0.7}{3}\right]\ (0.7 < M_0 \le 0.85)$$

$\eta_{prop,max}$ 0.80-0.90; chart $\eta_{prop} = \eta_L \eta_P$, no gearbox

$$F_{static} = (2\rho_0 A_D)^{1/3}(FM\,\dot{W}_{prop})^{2/3},$$

$$FM = \frac{F_{static}^{3/2}}{\sqrt{2\rho_0 A_D}\,\dot{W}_{prop}} \approx 0.70\text{-}0.80, \quad v_s = \sqrt{\frac{F}{2\rho_0 A_D}}$$

F_static ≈ W_prop^(2/3) A_D^(1/3)

$$F\big|_{M_0=0} = \frac{10\eta_{\max}\dot{W}_{prop}}{a_0}, \qquad FM_{implied} = \left(\frac{10\eta_{\max}}{a_0}\right)^{3/2}\sqrt{\frac{\dot{W}_{prop}}{2\rho_0 A_D}}$$

FM_implied ≤ 1 within the model; measured static FM 0.70-0.80

$$\dot{W}_{prop} = \eta_g(\eta_{m,L}\dot{W}_{LPT} - \dot{W}_{LPC}),$$

$$F_{prop} = \frac{\eta_{prop}\dot{W}_{prop}}{v_0}, \quad \dot{W}_D = (1 + a)\eta_{prop}\dot{W}_{prop}$$

$$\beta = \frac{\dot{m}_{prop}}{\dot{m}_{jet}}, \ \dot{m}_{jet} = \frac{\dot{m}_0}{1 + \beta},$$

$$\frac{\dot{W}_{prop}}{\dot{m}_0} = \frac{\eta_g c_{pc} T_0}{1 + \beta}\left[\eta_{mL}(1 + f)\tau_\lambda \tau_{tH}(1 - \tau_{tL}) - \tau_r(\tau_{cL} - 1)\right]$$

$\tau_{tH}$ from the HP balance with $\tau_r$ $\tau_{cL}$ ($\tau_{cH}$ - 1) on top

$$v_0 \frac{F}{\dot{m}_0} = \eta_{prop}\frac{\dot{W}_{prop}}{\dot{m}_0} + \frac{\gamma_c - 1}{1 + \beta} M_0 \left[(1 + f)\frac{v_9}{a_0} - M_0\right] c_{pc} T_0$$

$$\dot{m}_{prop} = \rho_0 A_D v_p, \quad \Delta E_k = \tfrac{1}{2}\dot{m}_9 v_9^2 - \tfrac{1}{2}\dot{m}_{jet} v_0^2 + \dot{W}_D,$$

$$\mathrm{TSFC} = \frac{f}{(1 + \beta)F/\dot{m}_0}, \quad \mathrm{PSFC} = \frac{f}{(1 + \beta)\dot{W}_{prop}/\dot{m}_0}$$

Mattingly uses W_prop instead of W_D: same $\eta_O$, lower $\eta_P$ and higher $\eta_T$

$$J = \frac{v_0}{ND}, \quad C_P = \frac{\dot{W}_{prop}}{\rho_0 N^3 D^5}, \quad \eta_{prop} = \frac{C_F}{C_P} J, \quad M_{tip} = \frac{\sqrt{v_0^2 + (\pi N D)^2}}{a_0}$$

C_P = 1.76327e8 W[kW]/(σ N[rpm]^3 D^5); drag divergence near M0 0.7, swept thin blades reach ≈ 0.8

$$\tau_{tL}^* = \frac{\tau_{tH}^{-1/e_{tH}}}{\Pi}\tau_{tL}^{-\frac{1 - e_{tL}}{e_{tL}}} + A\left[1 + \frac{1 - e_{tL}}{e_{tL}}\frac{\tau_{tH}^{-1/e_{tH}}\tau_{tL}^{-1/e_{tL}}}{\Pi}\right]^2$$

$$A = \frac{(\gamma_c - 1)M_0^2/2}{\tau_\lambda \tau_{tH}(\eta_{prop}\eta_g\eta_{mL})^2}, \qquad \Pi = (\pi_r \pi_d \pi_{cL} \pi_{cH} \pi_b \pi_n)^{\frac{\gamma_t - 1}{\gamma_t}}$$

faster flight or worse propeller → larger $\tau_{tL}$*: more energy left in the jet

$$\alpha_1 = \frac{\dot{m}_{prop}}{\dot{m}_{jet}}, \quad \alpha_2 = \frac{\dot{m}_{sec}}{\dot{m}_{core}},$$

$$\mathrm{TSFC} = \frac{f}{(1 + \alpha_1)(1 + \alpha_2)F/\dot{m}_0}, \quad A_D = \frac{\pi D^2}{4} - \dot{m}_{45}\frac{\sqrt{RT_{t45}}}{P_{t45}\mathrm{MFP}_{45}}$$

tractor propeller: A_D ≈ π D^2/4 (core streamtube neglected)

## Engine sizing

$$\dot{m}_0 = \frac{F_{inst} + D_{nac} + D_{add}}{F/\dot{m}_0}, \qquad A_i = \frac{\dot{m}_i\sqrt{R_i T_{ti}}}{P_{ti}\,\mathrm{MFP}(\gamma_i, M_i)}$$

$\dot{m}_i$: $\dot{m}_0$ at 0-3; (1+f) $\dot{m}_0$/(1+α) at core 4-9; α $\dot{m}_0$/(1+α) at fan 13-19; (1+f+α) $\dot{m}_0$/(1+α) at 6A-9 (mixed). R_c, $\gamma_c$ cold, R_t, $\gamma_t$ hot. C-D nozzle: throat 8 and exit 9 are different areas.

## Turbomachinery: Euler and conventions

$$\tau_A = \dot{m}(r_e v_e - r_i v_i), \qquad h_{te} - h_{ti} = \omega(r_e v_e - r_i v_i)$$

$$\text{compressor: } c_p \Delta T_t = U(v_2 - v_1),$$

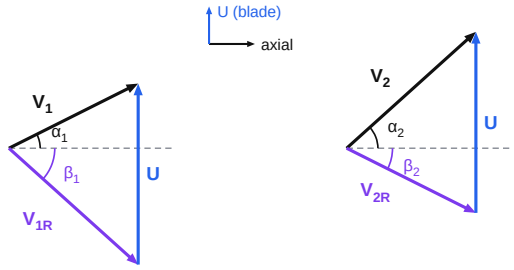
$$\text{turbine: } c_p \Delta T_t = U(v_2 + v_3), \quad U = \omega r_m$$

$$\vec{V} = \vec{V}_R + \vec{U}, \quad v = u\tan\alpha, \quad v_R = u\tan\beta, \quad r_m = \frac{r_t + r_h}{2}, \quad \omega = \frac{2\pi N}{60}$$

compressor: rotor 1-2, stator 2-3, v + v_R = U; turbine: nozzle 1-2, rotor 2-3, v2R = v2 - U, v3R = v3 + U (v3 positive opposite to U)

P, T, ρ are frame-invariant; Tt, Pt, M are not. Rotor: Tt,R constant (same radius). Stator: Tt constant.

## Compressor stage




**rotor inlet (1)** **rotor exit (2) = stator inlet**

angles from the axial direction; V = V_R + U, compressor v + v_R = U; rotor 1-2, stator 2-3; repeating stage: $\alpha_1 = \beta_2$, $\beta_1 = \alpha_2$

$$c_p\Delta T_t = Uu(\tan\beta_1 - \tan\beta_2) = Uu(\tan\alpha_2 - \tan\alpha_1)$$

$$T_{t1R} = T_1 + \frac{V_{1R}^2}{2c_p} = T_{t2R}, \quad P_{t2R} = L_R P_{t1R},$$

$$P_{t3} = L_S P_{t2}, \quad L_R = 1 - \phi_{cr}\frac{\gamma P_1 M_{1R}^2}{2P_{t1R}}$$

$$R_c = \frac{T_2 - T_1}{T_3 - T_1}, \quad \Psi = \frac{c_p\Delta T_t}{U^2},$$

$$\Phi = \frac{u_1}{U} = \frac{1}{\tan\alpha_1 + \tan\beta_1}, \quad \frac{\Psi}{\Phi} = \tan\beta_1 - \frac{u_2}{u_1}\tan\beta_2$$

$$\eta_{cs} = \frac{\pi_{cs}^{\varepsilon} - 1}{\tau_{cs} - 1}, \qquad e_{cs} = \frac{\gamma - 1}{\gamma}\frac{\ln(P_{t3}/P_{t1})}{\ln(T_{t3}/T_{t1})}, \qquad M_b = \frac{U}{a_1}$$

$$D = 1 - \frac{V_e}{V_i} + \frac{|v_i - v_e|}{2\sigma V_i} \le 0.6,$$

$$D_r = 1 - \frac{V_{2R}}{V_{1R}} + \frac{|v_{1R} - v_{2R}|}{2\sigma V_{1R}}, \quad D_s = 1 - \frac{V_3}{V_2} + \frac{|v_2 - v_3|}{2\sigma V_2}$$

de Haller Ve/Vi ≥ 0.72 (rough)

$$\sigma = \frac{c}{s}, \quad s = \frac{2\pi r_m}{n_b}, \quad i = \alpha_i - \gamma_i, \quad \delta = \alpha_e - \gamma_e, \quad \delta_c = \frac{\gamma_i - \gamma_e}{4\sqrt{\sigma}}, \quad \delta_t = \frac{\gamma_i + \gamma_e}{8\sqrt{\sigma}}$$

$$\gamma_e = \frac{4\alpha_e\sqrt{\sigma} - \gamma_i}{4\sqrt{\sigma} - 1}$$

$$\text{repeating stage: } v_1 + v_2 = U, \quad c_p\Delta T_t = V_2^2 - V_1^2,$$

$$\frac{\omega r}{V_1} = \cos\alpha_1(\tan\alpha_1 + \tan\alpha_2)$$

V1R = V2, V2R = V1 (symmetric triangles); T3 = Tt3 - V1^2/(2cp)

$$\tau_{cs} = 1 + \frac{(\gamma - 1)M_1^2}{1 + \frac{\gamma-1}{2}M_1^2}\left(\frac{\cos^2\alpha_1}{\cos^2\alpha_2} - 1\right),$$

$$\Psi = \frac{\tan\alpha_2 - \tan\alpha_1}{\tan\alpha_1 + \tan\alpha_2}, \quad \Phi = \frac{1}{\tan\alpha_1 + \tan\alpha_2}$$

$$D = 1 - \frac{\cos\alpha_2}{\cos\alpha_1} + \frac{\tan\alpha_2 - \tan\alpha_1}{2\sigma}\cos\alpha_2, \qquad M_3 = M_1\sqrt{\frac{T_1}{T_3}}, \qquad \pi_{cs} = \tau_{cs}^{\frac{\gamma e_c}{\gamma - 1}}$$

$$\pi_c = \left[\frac{T_{t1} + n_s\Delta T_t}{T_{t1}}\right]^{\frac{\gamma e_c}{\gamma - 1}},$$

$$n_s \approx \frac{T_{t1}\left(\pi_c^{(\gamma-1)/(\gamma e_c)} - 1\right)}{\Delta T_{t,\text{stage}}}, \quad M_{3,n} = M_1\sqrt{\frac{T_1}{T_1 + n_s\Delta T_t}}$$

stage ratios fall along the machine: $\pi_c \neq \pi_{cs}$^n; first stage set by inlet guide vanes

$$\frac{dh_t}{dr} = u\frac{du}{dr} + v\frac{dv}{dr} + \frac{v^2}{r}, \qquad \text{free vortex: } R = 1 - (1 - R_m)\left(\frac{r_m}{r}\right)^2$$

reaction lowest at the hub (turbine hub R_t typically > -0.15)

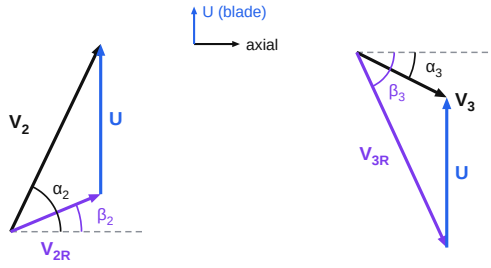
$$A = \frac{\dot{m}\sqrt{RT_t}}{P_t\cos\alpha\,\text{MFP}(M)}, \quad r_t = \sqrt{\frac{A}{\pi\left[1 - (r_h/r_t)^2\right]}},$$

$$h = \frac{A}{2\pi r_m}, \quad \omega = \frac{U_t}{r_t}, \quad N = \frac{30\omega}{\pi}$$

hub/tip 0.60-0.75 at HPC inlet, 0.90-0.92 at exit; Pt in Pa

| Compressor design range | Fan / LPC | HPC |
|---|---|---|
| pressure ratio: 1 / 2 / 3 stages | 1.5-2.0 / 2.0-3.5 / 3.5-4.5 | - |
| inlet corrected ṁ per area, kg/(s m^2) | 195-205 | 175-185 |
| tip speed (m/s) | 427-457 | rim at exit 396-457 |
| diffusion factor D | 0.50-0.55 | 0.50-0.55 |
| Φ / Ψ | - | 0.45-0.55 / 0.30-0.35 |
| first stage: hub/tip, axial M, U_tip | 0.6-0.75, 0.48-0.6 | 350-460 m/s |

## Turbine stage




**nozzle exit (2) = rotor inlet** **rotor exit (3)**

nozzle 1-2, rotor 2-3; v3 and $\beta_3$ positive opposite to U: v2R = v2 - U, v3R = v3 + U, Ψ = (v2 + v3)/U; $\alpha_3 = 0$ gives V3 = u

$$T_{t2} = T_{t1}, \quad P_{t2} = L_S P_{t1}, \quad T_{t3R} = T_{t2R}, \quad P_{t3R} = L_R P_{t2R},$$

$$L = \frac{1}{1 + \phi_t\left[1 - (T/T_t)^{\frac{\gamma}{\gamma-1}}\right]}$$

$$\Psi = \frac{c_p\Delta T_t}{U^2} = \frac{u_2}{U}\left(\tan\alpha_2 + \frac{u_3}{u_2}\tan\alpha_3\right) = \frac{u_2}{U}\left(\tan\beta_2 + \frac{u_3}{u_2}\tan\beta_3\right),$$

$$\Phi = \frac{u_2}{U} = \frac{1}{\tan\alpha_2 - \tan\beta_2}$$

$$\text{const. } u: \ \frac{\Psi}{\Phi} = \tan\beta_2 + \tan\beta_3, \quad \frac{R_t}{\Phi} = \frac{\tan\beta_3 - \tan\beta_2}{2}, \quad R_t = \frac{T_2 - T_3}{T_{t1} - T_{t3}}$$

$$R_t = 0: \ \Psi = 2\Phi\tan\beta_2 \ (\beta_3 = \beta_2);$$

$$R_t = 0.5: \ \Psi = 2\Phi\tan\alpha_2 - 1 \ (\beta_3 = \alpha_2); \quad \alpha_3 = 0: \ \Psi = 2(1 - R_t)$$

zero swirl + 0 %: Ψ = 2, v2 = 2U, v2R = v3R = U; zero swirl + 50 %: Ψ = 1; Ψ > 2 means R_t < 0

$$\Phi = \frac{1}{\tan\alpha_2 - \tan\beta_2}, \qquad U = \sqrt{\frac{c_p\Delta T_t}{\Psi}}, \qquad u = \Phi U$$

map axes: turning $\beta_2 + \beta_3$ vs $\beta_2$; usual limits $\alpha_2 \le 70°$ and Z 0.8-1.0

$$\eta_{ts} = \frac{1 - T_{t3}/T_{t1}}{1 - (P_{t3}/P_{t1})^{(\gamma-1)/\gamma}}, \qquad e_{ts} = \frac{\gamma}{\gamma - 1}\frac{\ln(T_{t3}/T_{t1})}{\ln(P_{t3}/P_{t1})}$$

$$\alpha_3 = 0: \ \sin\alpha_2 = \frac{\Psi U}{V_2}, \quad R_t = 1 - \frac{1}{2\Psi}\left(\frac{V_2}{U}\right)^2\left[1 - \left(\frac{V_3}{V_2}\right)^2\right],$$

$$T_3 = T_2 - R_t(T_{t1} - T_{t3}), \quad M_3 = M_2\frac{V_3}{V_2}\sqrt{\frac{T_2}{T_3}}$$

$$M_2\sin\alpha_2 = 2M_T(1 - R_t), \ M_T = \frac{U}{a_2};$$

$$M_{2,opt} = \sqrt{\frac{2}{\gamma - 1}\left[\left(\frac{2\left(1 + \frac{\gamma-1}{2}M_1^2\right)^{\frac{\gamma+1}{\gamma-1}}}{(\gamma + 1)M_1^2}\right)^{\frac{\gamma-1}{2}} - 1\right]}$$

$$Z = \frac{F_t}{F_{t\max}} = \frac{2s}{c_x}\cos^2\alpha_e\left(\tan\alpha_i + \frac{u_e}{u_i}\tan\alpha_e\right)\left(\frac{u_i}{u_e}\right)^2,$$

$$Z_r = 2\frac{s}{c_x}\cos^2\beta_3(\tan\beta_2 + \tan\beta_3)$$

design Z 0.8-1.0; equivalent loading parameter Z_r c_x/s

$$F_t = \rho u_i s(v_i + v_e) = \rho u_i^2 s\left(\tan\alpha_i + \frac{u_e}{u_i}\tan\alpha_e\right),$$

$$F_{t\max} = (P_{ti} - P_e)c_x = \frac{\rho u_e^2 c_x}{2\cos^2\alpha_e}$$

$$\sigma_c \approx \frac{\rho\omega^2 A}{4\pi}\left(1 + \frac{A_t}{A_h}\right) \ \Leftrightarrow \ AN^2 = \frac{3600 \times 10^3}{\pi(1 + A_t/A_h)}\frac{\sigma_c}{\rho}$$

N in rpm, A in m^2, $\sigma_c/\rho$ in MPa/(g/cm^3) = 1000 m^2/s^2; reduced by lower N, more taper or more stages

$$\mathrm{DSF} = \frac{(\omega r_r)^2 \rho}{2\sigma_d}, \qquad \sigma_{tr} = \frac{\alpha E \Delta T}{3}\left(1 - \frac{r}{r_h}\right), \quad \sigma_{t\theta} = \frac{\alpha E \Delta T}{3}\left(1 - \frac{2r}{r_h}\right)$$

cooled blades allow A N^2 ≈ 4-5e10 in^2 rpm^2

$$T_{t3}^{(n)} = T_{t1}^{(1)} - n_s \Delta T_t$$

only the first nozzle chokes and meters the engine flow

Choking: first nozzle M2 > 1 and rotor exit M3R < 0.9 (a 50 % first stage, Ψ = 1, cannot meter the flow). r_m = U/ω, h = A/(2 π r_m); typical LPT U 350-460 m/s.

## Two-spool turbofan and propfan

$$T_{t45} = T_{t4} - \frac{c_{pc}(T_{t3} - T_{t25})}{\eta_{mH}(1+f)c_{pt}},$$

$$T_{t5} = T_{t45} - \frac{c_{pc}\left[(1+\alpha)(T_{t13} - T_{t2}) + (T_{t25} - T_{t13})\right]}{\eta_{mL}(1+f)c_{pt}}$$

Propeller on a turbofan core (propfan, open rotor): $\dot m_{jet} = \dot m_{core}$; $\tau_{LPT} = 1 - \eta_{LPT}\,(1 - \pi_{LPT}^{\varepsilon})$; W_LPT = (1+f) $\dot m_{core}$ cp_t Tt45 (1 - $\tau_{LPT}$); W_prop = $\eta_g$ ($\eta_{mL}$ W_LPT - W_LPC); F_jet = $\dot m_9$ v9 - $\dot m_{core}$ v0 + (P9 - P0) A9.

$$\text{flight: } F_{prop} = \frac{\eta_{prop}\dot W_{prop}}{v_0},\ \dot W_D = F_{prop} v_p,\ \eta_L = \frac{\dot W_D}{\dot W_{prop}};$$

$$\text{static: } F_{prop} = (2\rho_0 A_D)^{1/3}(F M \dot W_{prop})^{2/3},\ v_p = \sqrt{\frac{F_{prop}}{2\rho_0 A_D}}$$

tractor propeller: A_D = π D^2/4; pusher or open rotor: annulus outside the nacelle

$$\dot m_{prop} = \rho_0 A_D v_p, \quad \text{bypass} = \frac{\dot m_{prop}}{\dot m_{core}},$$

$$\dot m_0 = \dot m_{core} + \dot m_{prop}, \quad \Delta E_k = \tfrac{1}{2}\left(\dot m_9 v_9^2 - \dot m_{core} v_0^2\right) + \dot W_D$$

## Repeating compressor stage (R = 0.5)

U = ω r_m; design chart: ω r/V1 and $\alpha_2$ as functions of $\alpha_1$ (M1 0.4-0.6). Rotor in the relative frame, stator in the absolute frame. γ = 1.4: γ/(γ-1) = 3.5, (γ-1)/γ = 0.2857.

$$u = V_1 \cos\alpha_1, \quad \tan\alpha_2 = \frac{U}{u} - \tan\alpha_1, \quad V_2 = \frac{u}{\cos\alpha_2},$$

$$V_{1R} = V_2,\ V_{2R} = V_1, \quad \Delta T_t = \frac{V_2^2 - V_1^2}{c_p}$$

$$T_{t1R} = T_1 + \frac{V_2^2}{2c_p} = T_{t2R}, \quad T_2 = T_{t1R} - \frac{V_1^2}{2c_p},$$

$$T_{t2} = T_{t1} + \Delta T_t, \quad T_3 = T_{t2} - \frac{V_1^2}{2c_p}, \quad T_{t3R} = T_3 + \frac{V_2^2}{2c_p}$$

$$P_{t1R} = P_1\left(\frac{T_{t1R}}{T_1}\right)^{3.5}, \quad P_{t2R} = P_{t1R} - \phi_{cr}\tfrac{1}{2}\rho_1 V_2^2,$$

$$P_2 = P_{t2R}\left(\frac{T_2}{T_{t1R}}\right)^{3.5}, \quad P_{t2} = P_2\left(\frac{T_{t2}}{T_2}\right)^{3.5}$$

rho1 = P1/(R T1); same as L_R = 1 - φ γ P M^2/(2 Pt)

$$P_{t3} = P_{t2} - \phi_{cs}\tfrac{1}{2}\rho_2 V_2^2, \quad P_3 = P_{t3}\left(\frac{T_3}{T_{t2}}\right)^{3.5}, \quad \pi_{cs} = \frac{P_{t3}}{P_{t1}},$$

$$\eta_{cs} = \frac{\pi_{cs}^{0.2857} - 1}{\Delta T_t / T_{t1}}, \quad e_c = \frac{0.2857 \ln \pi_{cs}}{\ln(1 + \Delta T_t / T_{t1})}$$

rho2 = P2/(R T2); R_c = (T2 - T1)/(T3 - T1) = 0.5 for the repeating stage

$$\Delta T_t = \frac{T_{t1}}{n_s}\left(\pi_c^{\frac{\gamma-1}{\gamma e_c}} - 1\right), \quad \pi_{cs} = \left(1 + \frac{\Delta T_t}{T_{t1}}\right)^{\frac{\gamma e_c}{\gamma - 1}},$$

$$\eta_c = \frac{\pi_c^{0.2857} - 1}{n_s \Delta T_t / T_{t1}}, \quad T_{t,exit} = T_{t1} + n_s \Delta T_t$$

V2^2 = V1^2 + cp dTt; n stages: $\pi_c$ = (1 + n dTt/Tt1)^(γ e_c/(γ-1))

## Turbine stage with constant axial velocity

$$\alpha_3 = 0,\ u \text{ const.}: \ \tan\alpha_2 = \frac{\Psi}{\Phi}, \quad \tan\beta_2 = \frac{\Psi - 1}{\Phi}, \quad \tan\beta_3 = \frac{1}{\Phi}, \quad R_t = 1 - \frac{\Psi}{2}$$

50 %: Ψ = 1, $\beta_2 = 0$, $\beta_3 = \alpha_2$; 0 %: Ψ = 2, $\tan\beta_2 = \tan\beta_3 = 1/\Phi$, $\tan\alpha_2 = 2/\Phi$

$$V_2 = \frac{u}{\cos\alpha_2}, \quad V_{2R} = \frac{u}{\cos\beta_2}, \quad V_3 = u, \quad V_{3R} = \frac{u}{\cos\beta_3} = \sqrt{u^2 + U^2}, \quad v_2 = \Psi U$$

$$\frac{Z_r c_x}{s} = 2\cos^2\beta_3(\tan\beta_2 + \tan\beta_3) = 2\sin 2\beta_3\ (R_t = 0), \quad = \sin 2\beta_3\ (R_t = 0.5)$$

$$T_3 = T_{t3} - \frac{u^2}{2c_p},\ P_3 = P_{t3}\left(\frac{T_3}{T_{t3}}\right)^k,\ T_{t3R} = T_3 + \frac{V_{3R}^2}{2c_p} = T_{t2R},\ P_{t3R} = P_3\left(\frac{T_{t3R}}{T_3}\right)^k = L_R P_{t2R}$$

k = γ/(γ-1) = 4.333; R = 285.9

$$T_2 = T_{t1} - \frac{V_2^2}{2c_p}, \quad P_2 = P_{t2R}\left(\frac{T_2}{T_{t2R}}\right)^k,$$

$$P_{t2} = P_2\left(\frac{T_{t1}}{T_2}\right)^k, \quad L_S = \frac{P_{t2}}{P_{t1}}, \quad M_2 = \frac{V_2}{\sqrt{\gamma R T_2}}$$

Tt2R = T2 + V2R^2/(2cp) equals Tt3R; zero reaction: T3 = T2

$$A_2 = \frac{\dot m \sqrt{R T_{t1}}}{P_{t2}\cos\alpha_2\,\mathrm{MFP}(M_2)}, \quad \frac{\sigma_c}{\rho} = \frac{\pi(1 + A_t/A_h)AN^2}{3.6 \times 10^6}\left[\frac{\text{MPa}}{\text{g/cm}^3}\right], \quad r_m = \frac{U}{\omega}$$

## Component figures of merit (Mattingly)

| Component | FoM | Lvl 2 | Lvl 3 | Lvl 4 |
|---|---|---|---|---|
| diffuser (A / C) | $\pi_{d,max}$ | 0.95 / 0.90 | 0.98 / 0.94 | 0.995 / 0.96 |
| compressor / fan | e_c / e_f | 0.84 / 0.82 | 0.88 / 0.86 | 0.90 / 0.89 |
| burner | $\pi_b$ / $\eta_b$ | 0.92 / 0.94 | 0.94 / 0.99 | 0.95 / 0.999 |
| turbine uncooled / cooled | e_t | 0.85 / 0.83 | 0.89 / 0.87 | 0.90 / 0.89 |
| afterburner | $\pi_{AB}$ / $\eta_{AB}$ | 0.92 / 0.91 | 0.94 / 0.96 | 0.95 / 0.99 |
| nozzle (D / F) | $\pi_n$ | 0.97 / 0.93 | 0.98 / 0.95 | 0.995 / 0.97 |
| shaft (alone / + PTO) | $\eta_m$ | 0.97 / 0.92 | 0.99 / 0.95 | 0.995 / 0.97 |
| max Tt4 / Tt7 (K) | | 1390 / 1670 | 1780 / 2000 | 2000 / 2220 |

A = subsonic, engines in nacelles; C = supersonic, engine in airframe; D = fixed convergent, F = variable C-D nozzle. Level 3 ≈ 1985-2005 technology.

## Isentropic flow tables

| M | Tt/T 1.4 | Pt/P 1.4 | MFP 1.4 | A/A* 1.4 | Tt/T 1.3 | Pt/P 1.3 | MFP 1.3 |
|---|---|---|---|---|---|---|---|
| 0.10 | 1.0020 | 1.0070 | 0.1176 | 5.8218 | 1.0015 | 1.0065 | 0.1134 |
| 0.20 | 1.0080 | 1.0283 | 0.2311 | 2.9635 | 1.0060 | 1.0263 | 0.2229 |
| 0.30 | 1.0180 | 1.0644 | 0.3365 | 2.0351 | 1.0135 | 1.0598 | 0.3249 |
| 0.40 | 1.0320 | 1.1166 | 0.4306 | 1.5901 | 1.0240 | 1.1082 | 0.4164 |
| 0.50 | 1.0500 | 1.1862 | 0.5111 | 1.3398 | 1.0375 | 1.1730 | 0.4951 |
| 0.60 | 1.0720 | 1.2755 | 0.5763 | 1.1882 | 1.0540 | 1.2560 | 0.5592 |
| 0.70 | 1.0980 | 1.3871 | 0.6257 | 1.0944 | 1.0735 | 1.3598 | 0.6081 |
| 0.80 | 1.1280 | 1.5243 | 0.6595 | 1.0382 | 1.0960 | 1.4877 | 0.6419 |
| 0.90 | 1.1620 | 1.6913 | 0.6787 | 1.0089 | 1.1215 | 1.6436 | 0.6612 |
| 1.00 | 1.2000 | 1.8929 | 0.6847 | 1.0000 | 1.1500 | 1.8324 | 0.6673 |
| 1.20 | 1.2880 | 2.4250 | 0.6645 | 1.0304 | 1.2160 | 2.3337 | 0.6465 |
| 1.50 | 1.4500 | 3.6710 | 0.5822 | 1.1762 | 1.3375 | 3.5259 | 0.5610 |
| 2.00 | 1.8000 | 7.8244 | 0.4058 | 1.6875 | 1.6000 | 7.6651 | 0.3763 |

Columns for γ = 1.4 (cold) and 1.3 (hot). MFP = ṁ sqrt(R Tt)/(Pt A), dimensionless (R outside). Choked: Tt/T* = 1.2, Pt/P* = 1.893 (γ 1.4); 1.15, 1.832 (γ 1.3).

## ISA standard atmosphere

| h (km) | T (K) | P (bar) | ρ (kg/m^3) | σ | a (m/s) |
|---|---|---|---|---|---|
| 0 | 288.15 | 1.0132 | 1.2250 | 1.0000 | 340.3 |
| 1 | 281.65 | 0.8987 | 1.1117 | 0.9075 | 336.4 |
| 2 | 275.15 | 0.7950 | 1.0065 | 0.8216 | 332.5 |
| 3 | 268.65 | 0.7011 | 0.9091 | 0.7421 | 328.6 |
| 4 | 262.15 | 0.6164 | 0.8191 | 0.6687 | 324.6 |
| 5 | 255.65 | 0.5402 | 0.7361 | 0.6009 | 320.5 |
| 6 | 249.15 | 0.4718 | 0.6597 | 0.5385 | 316.4 |
| 7 | 242.65 | 0.4106 | 0.5895 | 0.4812 | 312.3 |
| 8 | 236.15 | 0.3560 | 0.5252 | 0.4287 | 308.1 |
| 9 | 229.65 | 0.3074 | 0.4664 | 0.3807 | 303.8 |
| 10 | 223.15 | 0.2644 | 0.4127 | 0.3369 | 299.5 |
| 11 | 216.65 | 0.2263 | 0.3639 | 0.2971 | 295.1 |
| 12 | 216.65 | 0.1933 | 0.3108 | 0.2537 | 295.1 |
| 13 | 216.65 | 0.1651 | 0.2655 | 0.2167 | 295.1 |

T = 288.15 - 6.5 h (h in km) up to 11 km, then 216.65 K; P = 1.01325 (T/288.15)^5.2559 bar below 11 km, P = 0.22632 exp(-0.1577 (h - 11)) above.

## Conventions

Mechanical losses are charged to the turbine (W_c = $\eta_m$ W_t). Stations 4-9 use $\gamma_t$, R_t, cp_t. Pt in Pa in MFP. Flow angles are measured from the axial direction; turbine v3 is positive opposite to U. Propeller efficiencies use the disc power W_D.